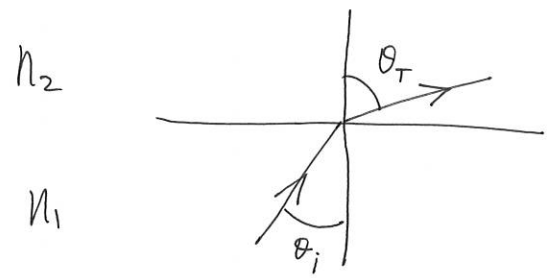
Total internal reflection $(n_1 > n_2)$

$$n_1 \sin\theta_i = n_2 \sin\theta_T$$



But $\sin\theta_T = \frac{n_1}{n_2}\sin\theta_i \to 1$ for $\sin\theta_{ic} = \frac{n_2}{n_1} < 1$:

at this $\theta_c = \theta_{ic}$ the refracted wave propagates at $\theta_T = \pi/2$

i.e. along the interface boundary.

For $\theta_i > \theta_c \longrightarrow$ total internal reflection.

Snell's law: $K_{2x} = K_{1x} = K_1 \sin\theta_i = \frac{\omega n_1}{c}\sin\theta_i$

$$\Rightarrow \quad K_{2z} = \sqrt{K_2^2 - K_{2x}^2} = \sqrt{\frac{\omega^2 n_2^2}{c^2} - \frac{\omega^2 n_1^2}{c^2}\sin^2\theta_i} = \frac{\omega n_2}{c}\sqrt{1 - \left(\frac{n_1 \sin\theta_i}{n_2}\right)^2}$$

$$= \frac{\omega n_2}{c}\sqrt{1 - \left(\frac{\sin\theta_i}{\sin\theta_c}\right)^2} = i\varkappa$$

$$\varkappa = \frac{\omega n_2}{c}\sqrt{\left(\frac{\sin\theta_i}{\sin\theta_c}\right)^2 - 1} > 0 \quad \text{for } \theta_i > \theta_c.$$

P-polariz. $\vec{E}_2(x,z) = \vec{E}_T(x,z) = \vec{E}_T\, e^{iK_{2x}x + iK_{2z}z - i\omega t}$

$$= \vec{E}_T\, e^{iK_{2x}x - i\omega t}\, e^{-\varkappa z}$$ decays exponentially into medium 2.

What are $\vec{E}_T$ components? $\vec{\nabla}\cdot\vec{E}_2(x,z) = 0$ $(\varepsilon_2 = \text{const})$

$$\Rightarrow \quad E_{Tx}\, iK_{2x} - E_{Tz}\,\varkappa = 0$$

$$\Rightarrow \quad \frac{E_{Tz}}{E_{Tx}} = \frac{iK_{2x}}{\varkappa} = \frac{iK_1 \sin\theta_1}{\varkappa} = i\,\frac{n_1 \sin\theta_i \sin\theta_c}{n_2\sqrt{\sin^2\theta_i - \sin^2\theta_c}}$$

Physical field $\vec{E}_2 = Re\left\{ \vec{E}_T(x,z) \right\} =$

$$= Re\left\{ \left(E_{Tx}\,\hat{x} + \frac{i k_1 \sin\theta_1}{\varkappa} E_{Tx}\,\hat{z} \right) e^{-\varkappa z}\, e^{i(k_{2x}x - \omega t)} \right\}$$

$$= E_{Tx}\, e^{-\varkappa z} \cos(k_{2x}x - \omega t)\,\hat{x} - \frac{k_1 \sin\theta_1}{\varkappa} E_{Tx}\, e^{-\varkappa z} \sin(k_{2x}x - \omega t)\,\hat{z}.$$

let $\mu_j = \mu_0$: $\quad r_p \equiv \frac{H_R}{H_i} = \frac{Z_1 \cos\theta_i - Z_2 \cos\theta_T}{Z_1 \cos\theta_i + Z_2 \cos\theta_T} \rightarrow \frac{n_2 \cos\theta_i - n_1 \cos\theta_T}{n_2 \cos\theta_i + n_1 \cos\theta_T}$

$$\left.\begin{aligned} k_{2z} &= k_2 \cos\theta_T \\ \cos\theta_T &= \frac{i\varkappa}{k_2} \end{aligned}\right\} \Rightarrow r_p = \frac{n_2 k_2 \cos\theta_i - i n_1 \varkappa}{n_2 k_2 \cos\theta_i + i n_1 \varkappa} = e^{-i\delta}.$$

$\Rightarrow R_p = |r_p|^2 = 1 \Rightarrow$ this is full reflection indeed.

Simply: $\cos\theta_T = \sqrt{1 - \sin^2\theta_T} = \sqrt{1 - \left(\frac{k_1}{k_2}\sin\theta_i\right)^2} = \sqrt{1 - \left(\frac{n_1 \sin\theta_i}{n_2}\right)^2}$

$k_2 \sin\theta_T = k_1 \sin\theta_i \quad , \quad k_1 = \frac{\omega}{v_1} = \frac{\omega n_1}{c}$

$k_2 = \frac{\omega}{v_2} = \frac{\omega n_2}{c}$

$$\cos\theta_T = \sqrt{1 - \left(\frac{\sin\theta_i}{\sin\theta_c}\right)^2} = \frac{i\varkappa}{k_2}$$